

Internal Structure and Grammatical Dynamics in Transformer-Based Language Generation

Introduction

Modern transformer architectures are capable of producing highly coherent language despite relying on probabilistic next-token prediction. The extent to which the internal generation process itself is structured, deterministic, or state-like remains unclear.

The experiments described in this document were designed to investigate the internal representation of grammatical structure during autoregressive text generation. The work builds upon earlier DFA-based analysis of part-of-speech (POS) transition structure in LLM-generated text, which revealed that grammatical generation tends to operate within a stable core of highly predictable transitions alongside regions of ambiguity and instability. A detailed report of that analysis can be found [here](#).

To further investigate those dynamics, a series of probing and entropy experiments were performed on hidden representations from GPT-2. The primary goals were the following:

1. Determine where grammatical structure becomes recoverable during generation.
2. Investigate whether grammatical uncertainty is explicitly encoded in hidden states.
3. Examine how long-range context influences grammatical continuation behavior.
4. Use these findings to motivate a hierarchical grammar-aware generation system

1. Next-POS Probe Experiments

The first set of experiments investigated whether the next grammatical continuation could be recovered from intermediate transformer representations.

To study this, linear probing classifiers were trained on hidden representations extracted from GPT-2 during next-token prediction. Two prediction tasks were evaluated:

1. Current POS prediction: predict POS_t from h_t
2. Next POS prediction: predict POS_{t+1} from h_t

where:

- h_t represents the hidden layer representation of the current token at a given transformer layer
- POS_t is the grammatical class of the current token

- POS_{t+1} is the grammatical class of the next token to be generated

The experiments used sentence-prefix contexts sampled from 995 English texts from Project Gutenberg. Project Gutenberg is a collection of more than 75,000 publicly available eBooks spanning a wide range of genres, writing styles, and historical periods. The same sentence-prefix contexts used in the earlier DFA analyses were used to maintain consistency between the structural graph analysis and the probing experiments.

For each context, the hidden states produced by GPT-2 were extracted from every transformer layer during inference. Several feature extraction strategies were evaluated to determine how grammatical information is distributed across the sequence representation:

- **Final-token representation:** uses only the hidden representation of the final token in the prefix.
- **Mean-pooled hidden states:** averages hidden representations across all tokens in the prefix
- **Last-k token pooling:** pools hidden representations over the final k tokens of the prefix
- **Concatenated layer representations:** concatenates hidden states from multiple neighboring layers

The probing classifiers were implemented using multinomial logistic regression models trained independently on each layer and feature representation.

Key Findings

Current POS is Highly Recoverable from Early Layers

One of the strongest findings from the probe experiments was that the current token's POS class could be predicted with extremely high accuracy from the earliest hidden layers of GPT-2.

Beginning at layer 0, current POS prediction accuracy already exceeds 90%. This is particularly striking because layer 0 contains no contextual reasoning, attention computation, or sequence interaction. The representation at this stage consists only of static token embeddings and positional information. This suggests that GPT-2 embeddings are already heavily organized around syntactic regularities. Words with similar grammatical behavior appear to cluster together in embedding space, meaning that part-of-speech structure is already partially linearly separable before contextual processing begins.

From there, current POS prediction accuracy improves only slightly across the network, peaking at approximately 93.8% around the middle layers before gradually declining to roughly 89% by the final layer. The relatively small improvement after layer 0 suggests that the network does not require deep computation to identify the grammatical role of the current token. Instead, the grammatical identity of the current word is largely encoded in the lexical embedding itself.

This finding suggests that early transformer layers primarily preserve and refine local lexical and syntactic identity rather than constructing grammatical continuation behavior.

Grammatical Trajectory is Strongly Recoverable in Middle Layers

Next-POS prediction behaves fundamentally differently from the current POS prediction. While current POS information is immediately recoverable from the earliest layers, next-POS prediction accuracy gradually improves as information propagates through the transformer stack. Prediction performance peaks in the middle of the network, particularly around layers 4–6, before gradually degrading toward the final layers. This suggests a layered computational organization inside the transformer.

The middle layers appear to organize predictive grammatical structure by integrating local syntactic information with contextual constraints from the surrounding sequence. These layers encode the most recoverable representation of grammatical continuation behavior.

The degradation in next-POS recoverability toward the final layers suggests that later transformer blocks progressively specialize toward, lexical discrimination, semantic precision, token-level competition, and output-logit shaping.

In other words, the network appears to construct grammatical continuation constraints in the middle layers before partially transforming or overwriting them during the final stages of lexical prediction.

The Final Token Functions as a Compressed Continuation State

Experiments using mean pooling and last-k pooling did not improve next-POS prediction performance relative to the final token representation. In fact, performance degraded substantially under both approaches.

Layer concatenation of the 4th, 5th, and 6th layers, in contrast, produced only marginal improvements over the single-layer final-token representation. Taken together, these findings suggest two important properties of transformer representations:

First, grammatical predictive structure is not distributed uniformly across the input sequence. Instead, the model progressively compresses the most useful continuation information into the final token representation.

Second, the middle-layer representations around layers 4–6 appear highly stable and structurally aligned. The marginal gains from concatenating neighboring layers suggest that these layers do not encode fundamentally different grammatical abstractions, but rather closely related representations of the same underlying continuation structure.

The results, therefore, support the idea that the final token behaves like a compressed

continuation state. The transformer progressively integrates sequence-level information into a compact representation used to drive future prediction.

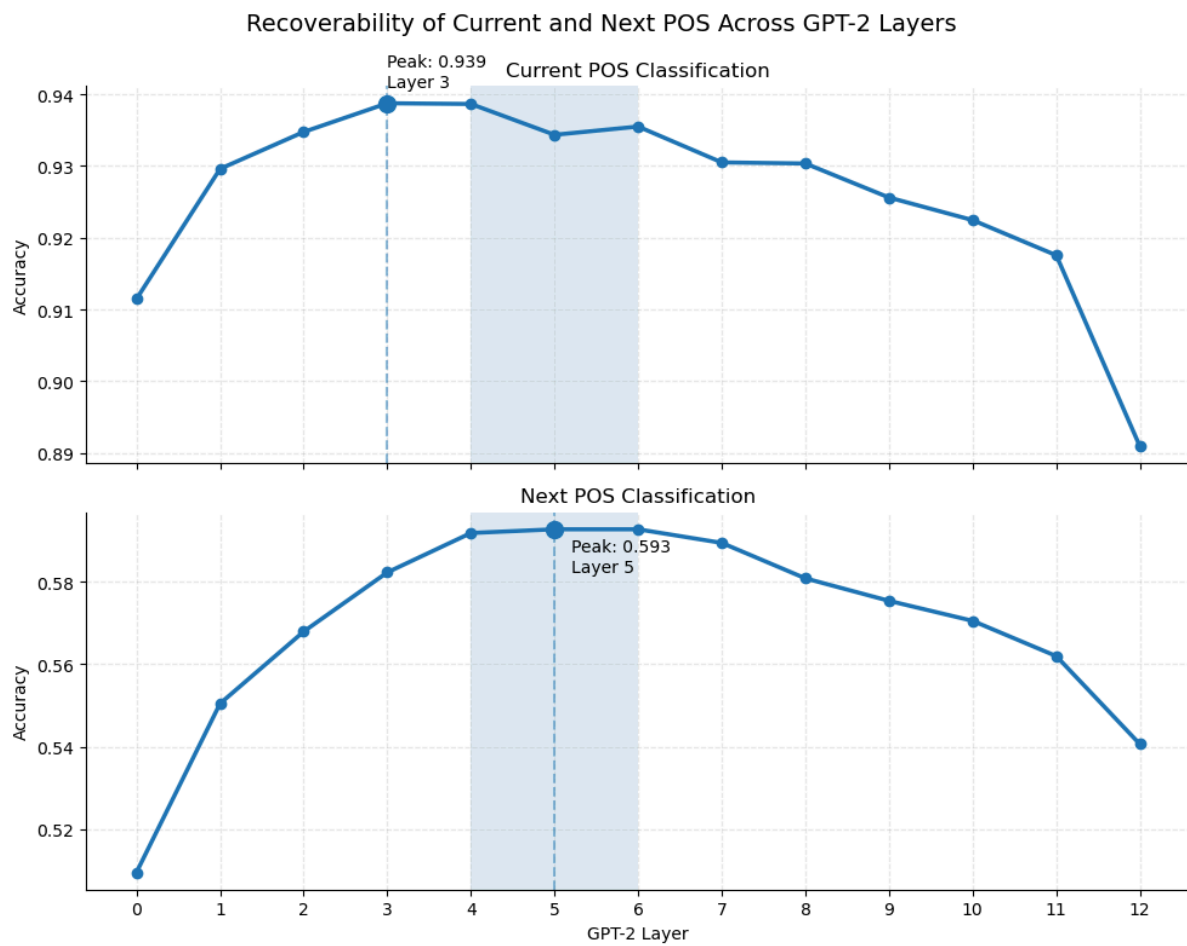


Figure 1: Layer-wise recoverability of current and next (POS) information across GPT-2 layers. Current POS remains highly recoverable throughout the network. Next-POS prediction peaks in the middle layers (Layers 4-6)

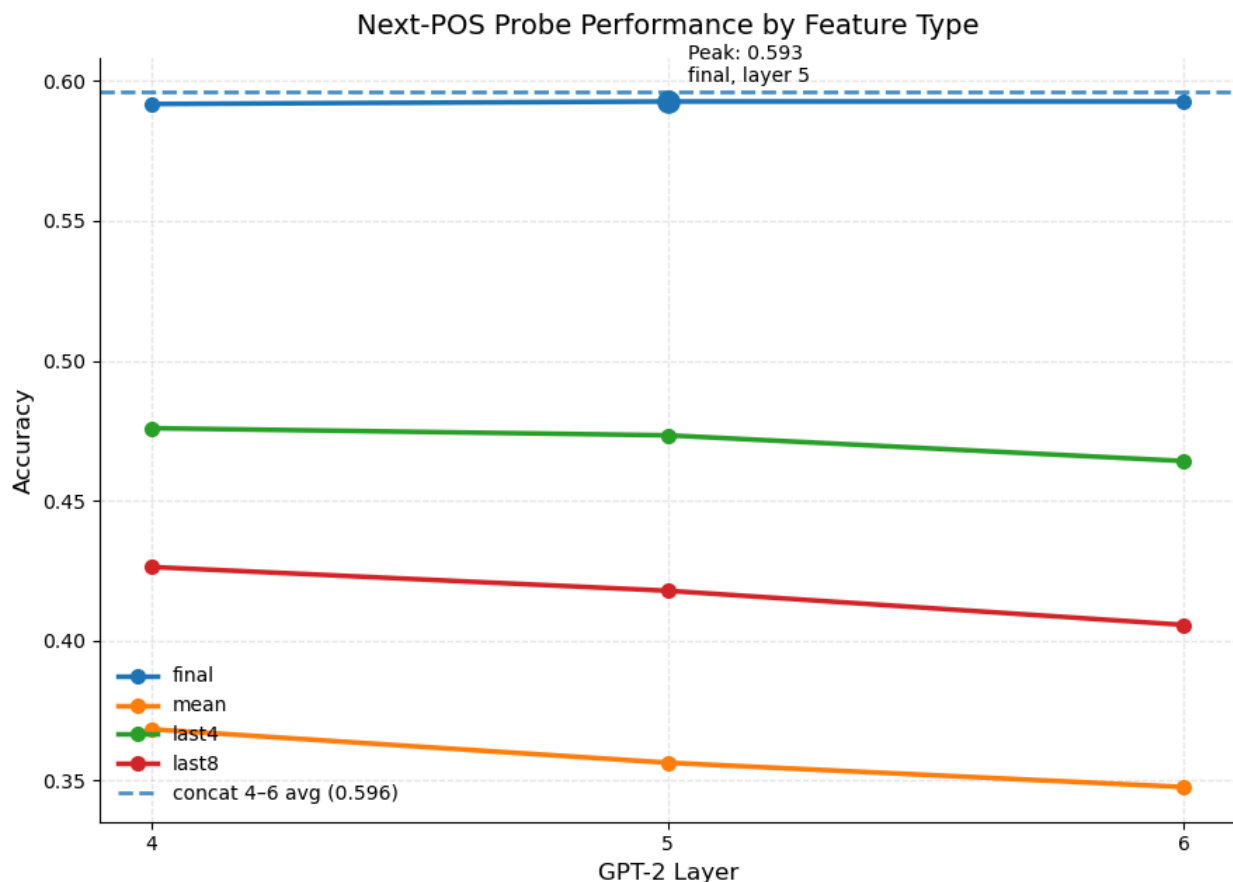


Figure 2: Next-POS probe performance across feature representations. The final-token hidden state consistently outperforms mean pooling and last-k pooling for next-POS prediction. Combining layers 4-6 through concatenation produces only a modest improvement over the best individual layer.

Structured Grammatical Topology Resides Within Transformer Representations

Analysis of the confusion matrices from the next-POS classification probes revealed that probe errors were highly structured rather than random. Misclassifications tended to remain within semantically adjacent, structurally adjacent, or grammatically neighboring regions of the POS space. This suggests that the hidden representations learned by the transformer encode an internal grammatical topology rather than isolated independent categories.

One strong example appears between adjectives (ADJ) and nouns (NOUN). A large proportion of adjective continuations were misclassified as nouns. This is linguistically meaningful because adjectives and nouns share overlapping distributional environments in English, particularly within noun-phrase constructions and modifier structures.

Many English words naturally function as both adjectives and nouns depending on context, including examples such as

- poor
- American
- criminal
- medical
- daily

Similarly, verbs (VERB) and auxiliaries (AUX) exhibited strong bidirectional confusion. Auxiliary verbs are structurally verb-like but semantically more constrained and highly recurrent. This likely explains why AUX continuations were more predictable overall while VERB continuations remained more ambiguous.

Additional neighboring grammatical relationships that emerged throughout the confusion analysis include

- Determiner (DET) and pronoun (PRON)
- Subordinating conjunction (SCONJ) and coordinating conjunctions (CCONJ)
- Symbol (SYM) and punctuation (PUNCT)

These findings suggest that grammatical categories inside transformer representations form overlapping manifolds rather than strictly isolated symbolic states.

Emergence of Higher-Level Grammatical Superclasses

The confusion matrix analysis further reveals the existence of broader latent grammatical superclasses inside the learned representation space. Several POS categories can be naturally grouped together based on shared confusion dynamics and contextual behavior:

Boundary / Closure Superclass

These categories govern formatting, segmentation, and sequence closure behavior:

- PUNCT
- SYM
- SPACE

Nominal Superclass

These classes participate heavily in noun-phrase construction and referential structure:

- NOUN
- PROPN
- PRON
- DET

Notably, NOUN was the single most predictable category across all experiments, suggesting that natural language generation strongly revolves around noun-phrase continuation dynamics.

Predicate Superclass

These categories govern predicate structure and clause progression.

- VERB
- AUX

Modifier Superclass

These classes modify entities, predicates, and local semantic structure:

- Adjective (ADJ)
- Adverb (ADV)

Connector Superclass

These categories govern grammatical linkage, dependency structure, and clause coordination:

- Adposition (ADP)
- CCONJ
- SCONJ
- Particle (PART)

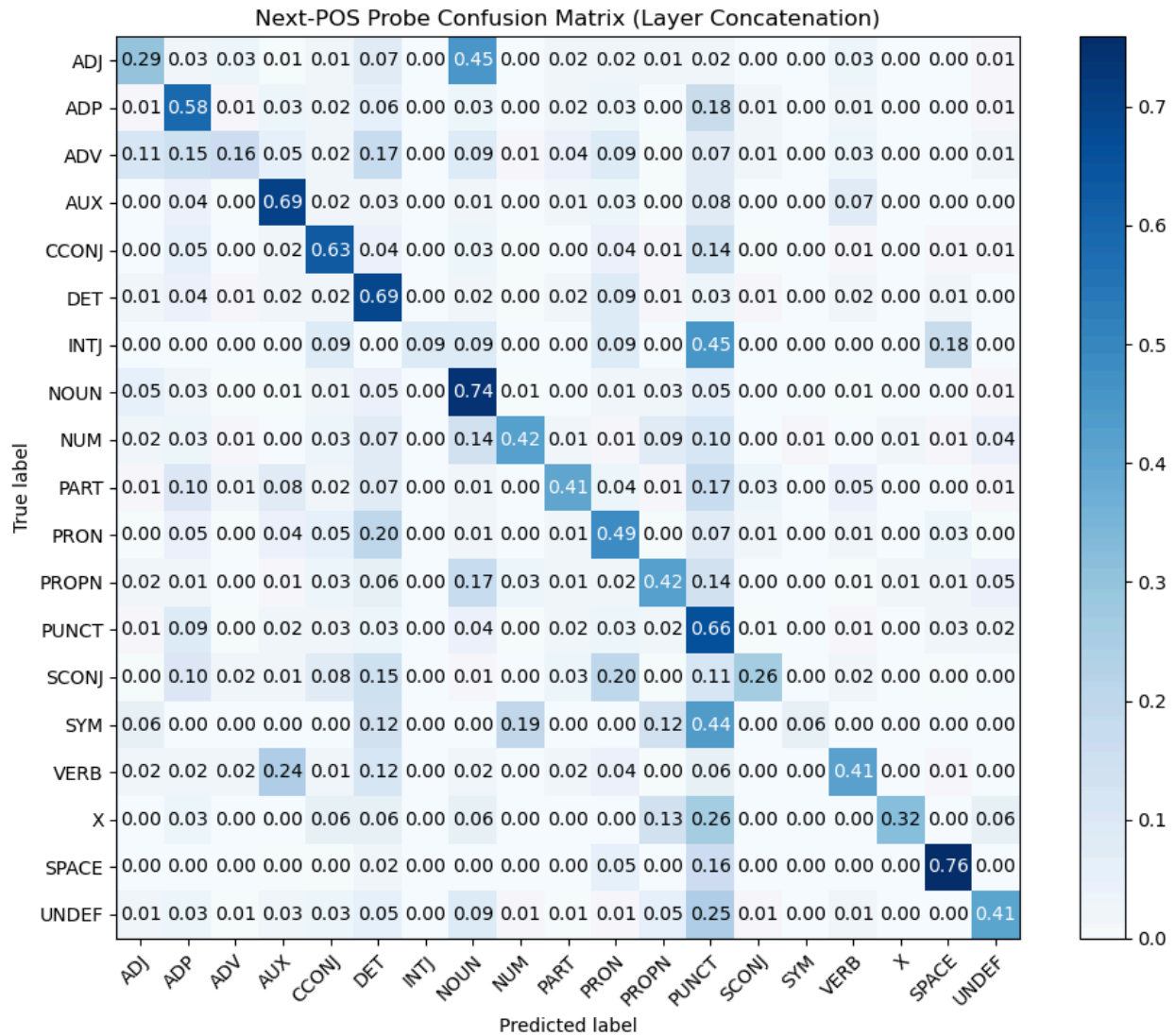


Figure 3: Next-POS confusion matrix from layer concatenated classification probe

2. Entropy Probe Experiments

The entropy probe experiments were motivated by observations from the earlier DFA analyses. Those experiments revealed that some grammatical states exhibit highly deterministic continuation behavior, while others contain broad branching structures and elevated uncertainty. Certain POS transition states consistently converged toward a dominant continuation, whereas others supported multiple competing grammatical continuations.

While the earlier next-POS probe experiments demonstrated that grammatical continuation structure is strongly recoverable from intermediate transformer layers, those experiments primarily focused on predicting the most likely grammatical continuation. The entropy experiments instead focused on the uncertainty surrounding that continuation.

The experiments ask:

“Does the transformer internally encode how uncertain the next grammatical continuation is?”

To investigate this question, regression probes were trained to predict next-POS entropy from hidden representations extracted from GPT-2 during autoregressive generation. Both linear and nonlinear regression probes were evaluated across multiple layers and feature representations.

Entropy Label Construction

For each sentence-prefix context, the top 100 next-token candidates were extracted from GPT-2 using the raw output logits under natural decoding conditions.

Importantly, no temperature scaling was applied:

temperature = 1.0

This means that the experiments measured the model’s natural next-token uncertainty distribution rather than an artificially sharpened or flattened decoding regime.

Each candidate token produced by the model was converted into a grammatical continuation using the same POS-tagging pipeline used in the DFA analyses. Candidate probabilities were then aggregated by POS class to construct a next-POS probability distribution.

Formally, the POS entropy for a context was defined as:

$$H(POS) = - \sum P(POS_i) \log P(POS_i)$$

Where $P(POS_i)$ represents the aggregated probability mass assigned to POS class i across the candidate continuation set. This formulation captures how uncertain the model is about the type of grammatical continuation rather than the specific next word.

Hidden-State Entropy Probes

Hidden states were extracted from every GPT-2 layer during inference for each sentence prefix. Because the earlier next-POS probe experiments demonstrated that the final token behaves like a compressed continuation representation, the primary feature representation used in the entropy probes was the final-token hidden state. The experiments therefore tested whether the hidden representation associated with the current token contains sufficient information to predict the uncertainty of the next grammatical continuation.

In addition to intermediate hidden states, a second type of probe was trained using logit-space features derived directly from the model's output distribution. The inclusion of these features was motivated by the hypothesis that grammatical uncertainty may not be explicitly encoded through the hidden-state trajectory but instead may merge during vocabulary competition and softmax normalization near the output stage of generation.

The extracted logit-space features included:

Top-k probability mass: Measures the cumulative probability mass captured by the top-k candidate tokens. Formally:

$$TopKMass(k) = \sum P(token_i)$$

For the top-k tokens ranked by probability. This feature captures how concentrated or diffuse the output distribution is. Highly concentrated distributions indicate strong preference toward a small number of candidate continuations, while diffuse distributions indicate broader uncertainty.

Token entropy: Measures entropy directly over the token-level probability distribution.

$$H(Token) = - \sum P(token_i) \log P(token_i)$$

Probability margins: Measures the difference between the highest-probability candidate and competing candidates. Included specifically are:

Top1 - Top2 margin, and

Top1 logit - Top2 logit

These features capture how decisively the model favors one continuation over alternatives.

Large margins indicate stable continuation behavior, while small margins indicate competitive ambiguity among candidate continuations.

Logit statistics: Statistical summaries computed directly from the raw logits, including:

- Mean,
- Standard deviation, and
- Maximum logit value

Full-vocabulary token entropy: Measures entropy across the full GPT-2 vocabulary distribution rather than only the truncated top-k candidate set.

Key Findings

One of the central findings from the entropy probe experiments was that exact next-POS entropy is largely unrecoverable from GPT-2 hidden representations. Across all layers and feature representations, both linear and nonlinear regression probes failed to meaningfully outperform a simple mean-baseline predictor. The strongest-performing models achieved errors that were nearly indistinguishable from baseline performance, while multilayer perceptron (MLP) probes consistently performed worse, indicating that increasing model capacity does not reveal a hidden nonlinear entropy signal.

The negative result was consistent across layers. No layer exhibited a clear entropy-representing "sweet spot," in contrast to the next-POS probing experiments where grammatical continuation information peaked in the middle layers. Likewise, logit-derived features and hidden-state features performed similarly, with neither providing substantial predictive power. Correlations between predicted and observed entropy remained extremely weak throughout.

Taken together, these results suggest that grammatical uncertainty is not encoded as a readily recoverable latent variable within GPT-2's hidden-state trajectory. Instead, uncertainty appears to emerge from the interaction of many weak signals during vocabulary competition and output decoding. While the network clearly encodes information about likely grammatical continuations, it does not appear to explicitly represent the uncertainty associated with those continuations in a form that can be easily extracted by either linear or nonlinear probes.

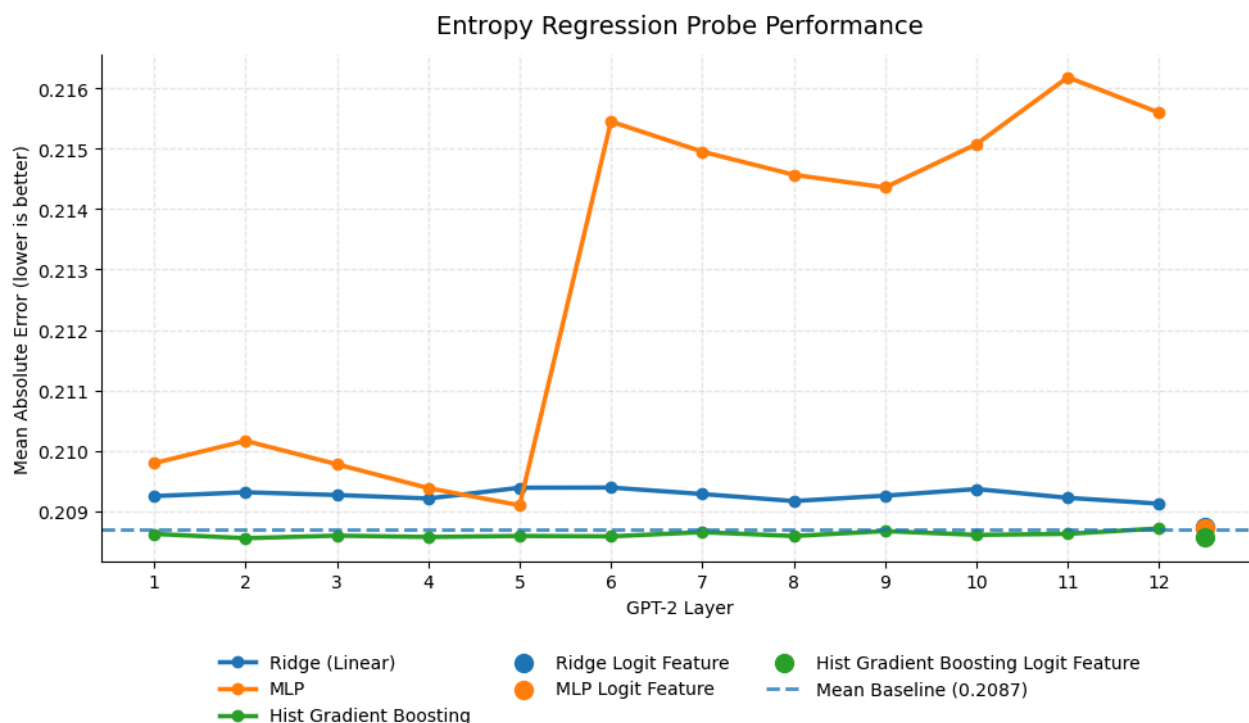


Figure 4: Entropy regression probe performance across multiple transformer layers in GPT-2. Both linear and nonlinear regression probes failed to meaningfully recover next-POS entropy.

3. Prefix-Length Context Experiments

The earlier DFA analyses primarily focused on local grammatical context, while the next-POS probe experiments demonstrated that the final token representation provides the strongest signal for predicting the next grammatical continuation. However, those probe experiments were performed primarily using sentence-prefix contexts, leaving an important question unresolved:

Does the final-token representation also compress useful long-range contextual information, or is it dominated primarily by local syntax?

The prefix-length context experiments were designed to investigate how increasingly long-range context influences grammatical continuation behavior, next-POS uncertainty, hidden-state organization, and the stability of POS distributions.

Context Formation

The experiments systematically varied the amount of preceding context while holding the local suffix constant. Starting from a local context size of 8 tokens, progressively larger contexts were constructed around the same suffix continuation. Approximately 10,000 suffixes were sampled from texts drawn from the Gutenberg corpus.

For each suffix, extended contexts were created at the following lengths:

- 16 tokens
- 32 tokens
- 64 tokens
- 128 tokens
- 256 tokens
- 512 tokens
- 768 tokens
- 1024 tokens

Most importantly, the suffix remained fixed across all context lengths. Only the amount of preceding context changed. This was done to isolate the effect of long-range contextual information on the model's grammatical continuation behavior while controlling for local syntactic structure. Contexts were aligned using the GPT-2 tokenizer to ensure that the extracted context windows accurately reflected the model's internal tokenization process.

POS Distribution and Entropy Analysis

For each context length, GPT-2 was run independently on the corresponding sequence, and the next-token logits were extracted. The top 100 candidate tokens were then converted into POS continuations using the same POS-tagging pipeline used in the DFA and entropy probe experiments.

Candidate probabilities were aggregated by POS class to construct next-POS probability distributions for each context length. The distributions are used to calculate the next-POS entropy for the input sequence.

Hidden-State Compression and Stability Analysis

To investigate how long-range context reorganizes internal transformer representations, hidden states from layers 4, 5, and 6 were extracted for every context length. These layers were selected because the earlier next-POS probe experiments demonstrated that grammatical continuation structure is most recoverable in the middle layers of the transformer stack. For each row, context length, and layer, cosine distance was computed between the final-token hidden representation at that context length and two reference contexts.

context_8 and context_1024

The reference contexts were also used to calculate stability metrics across different context lengths:

- **Top-POS agreement:** measures dominant grammatical continuations across different context lengths.
- **Jensen-Shannon Divergence:** Measures how the grammatical continuation landscape changes as additional context is introduced. Low divergence indicates stable grammatical behavior, while high divergence indicates that long-range context substantially reshapes the continuation distribution.
- **Entropy delta:** Changes in POS entropy across context lengths, measured relative to both context_8 and context_1024.

Key Findings

Long-Range Context Reshapes the Grammatical Continuation Landscape

Long-range context does not simply make the model “more confident” or “less confident” about the next grammatical continuation. Increasing context actively reshapes the grammatical continuation landscape itself.

The overall entropy profile revealed a particularly interesting pattern. As context length expanded from 8 to approximately 64 tokens, next-POS entropy initially increased before later stabilizing and declining under longer contexts. Very short contexts appear to constrain generation into relatively stereotypical grammatical trajectories. As more context is introduced, broader semantic and discourse-level possibilities become available, temporarily increasing grammatical branching and uncertainty. Only later does long-range contextual structure begin to stabilize the continuation landscape again. This behavior aligns closely with earlier DFA findings, which revealed localized regions of elevated grammatical branching and context-sensitive transition instability.

Additional evidence of contextual reshaping emerged through the analysis of the agreement and divergence metrics. Agreement with the short-context baseline (context_8) steadily decreased as longer contexts were introduced, while JS divergence from context_8 progressively increased. Agreement with the longest-context baseline (context_1024) also evolved gradually across intermediate context lengths, suggesting that grammatical continuation behavior continuously reorganizes as additional contextual information becomes available.

Taken together, findings suggest that semantic context influences grammar primarily through distributional shaping. For example, longer context may not directly force a transition such as VERB -> NOUN but instead continuously alters the continuation probabilities.

P(VERB)
P(NOUN)
P(ADJ)
P(PRON)
...

Long-range contexts appear to reshape which grammatical futures the model considers plausible during generation.

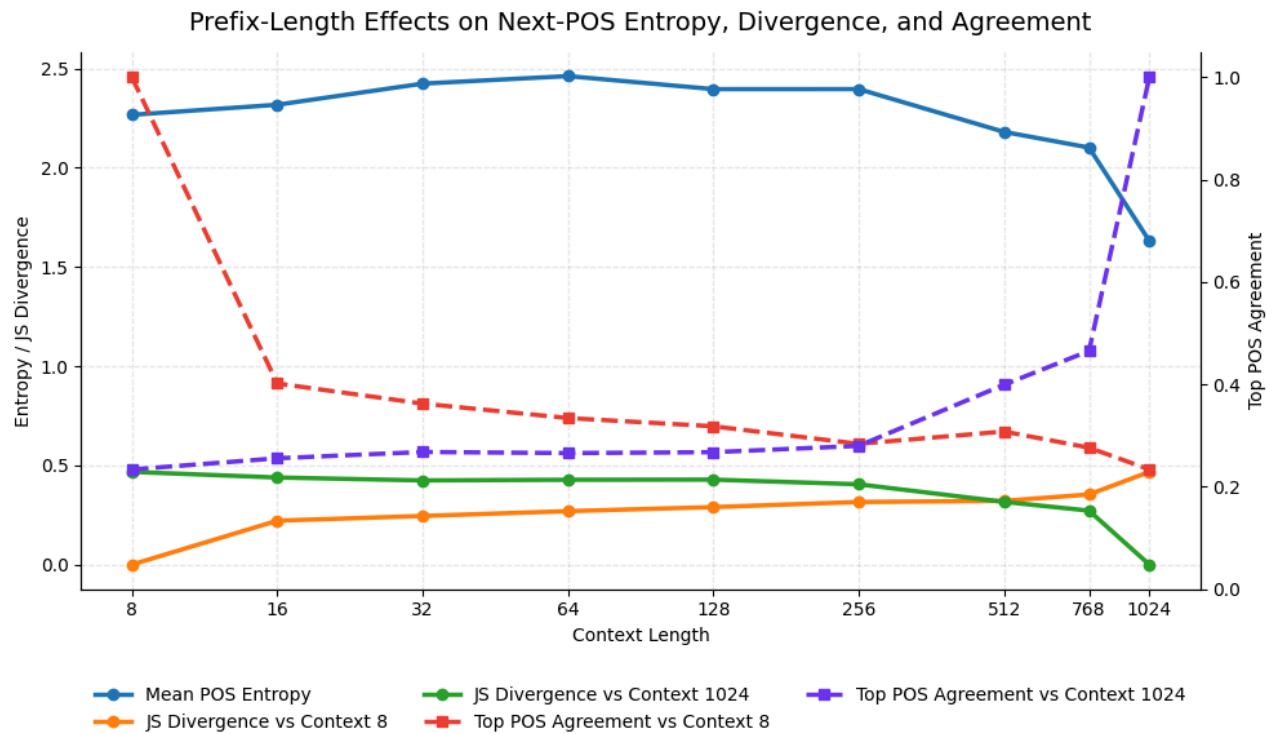


Figure 5: Effects of input prefix length on next-POS entropy, divergence, and POS agreement.
Average entropy across all sequences shows a meaningful increase as more context is introduced before declining under long contexts.

Final-Token Representations Progressively Integrate Long-Range Context

The hidden-state distance analysis revealed that the cosine distance between final-token representations consistently increased as context length increased. This demonstrates that long-range context progressively reorganizes the hidden representation associated with the same local suffix continuation.

Most importantly, the magnitude of this contextual deformation varied substantially across layers. The increase in hidden-state distance was weakest in layer 4 and strongest in layer 6. This suggests that deeper transformer layers gradually integrate and reorganize long-range contextual information into the final token representation. It also implies a hierarchical organization of contextual processing:

- In layer 4, representations remain relatively stable and locally anchored, preserving short-range grammatical structure.
- In layer 5, intermediate contextual information becomes increasingly integrated into the final token representation
- In layer 6, the strongest long-range contextual reorganization emerges

The results support one of the central findings from the earlier next-POS probe experiments: the final token behaves like a compressed summary of the preceding context.

This behavior closely parallels the dynamics of modern State Space Models (SSMs), including architectures such as Mamba. In SSMs, sequence history is recursively compressed into a latent state that evolves over time according to structured state updates:

$$h_t = Ah_{t-1} + Bx_t$$

The prefix-length experiments suggest that transformers may exhibit a similar form of latent state evolution. Rather than storing contextual information explicitly across all prior tokens equally, the network appears to progressively compress the long-range information into the final-token representation used for future prediction.

This interpretation aligns with a [recent theoretical work connecting attention mechanisms and State Space Models](#). Gu and Dao argued that attention and SSMs should not necessarily be viewed as completely separate paradigms. Under certain formulations, both can be interpreted as mechanisms for propagating contextual information through sequences via structured state updates.

The experiments suggest that transformer generation may internally behave more like a latent state-evolution system than a purely token-level probabilistic mechanism. The hidden-state trajectory appears to progressively accumulate and compress contextual structure over time, while probabilistic uncertainty primarily emerges later during vocabulary-level decoding.

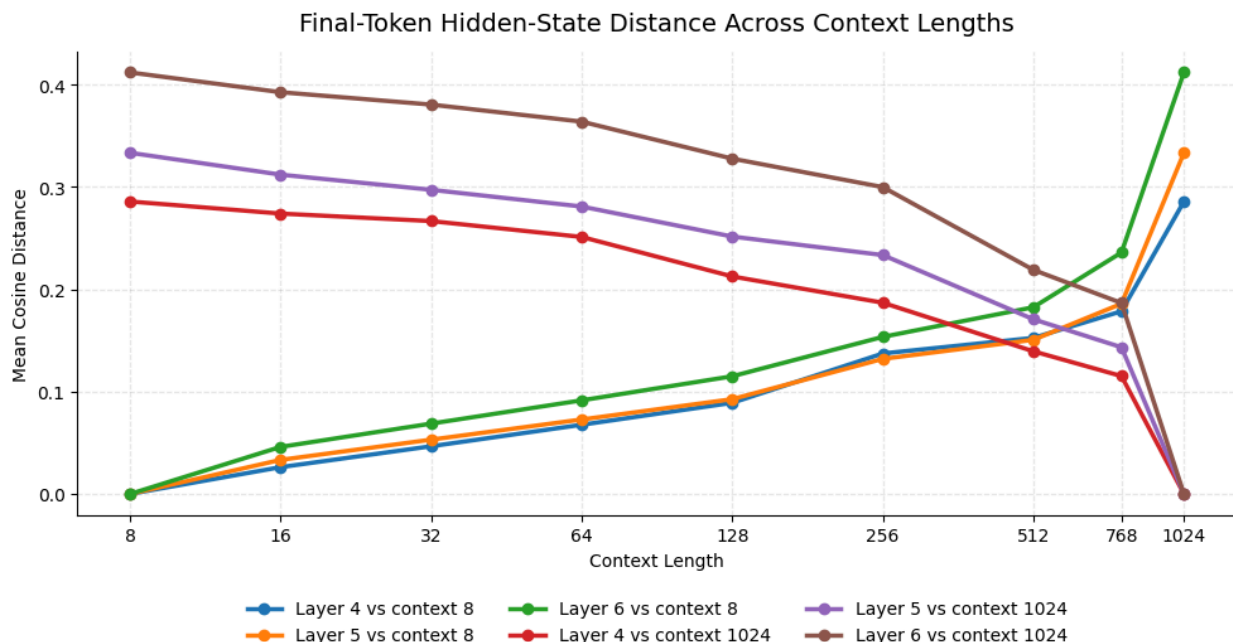


Figure 6: Hidden state cosine distance across context lengths in the middle transformer layers.
The final token appears to serve as a compressed summary of the preceding context, as is evident by the increase in cosine distance from the reference baseline over longer contexts.

Grammar Classes Separate Into Distinct Context-Sensitive Regimes

Analysis of how entropy, agreement, and divergence evolve for each POS category under increasing context length revealed that grammatical classes separate into three broad context-sensitive regimes. These regimes align closely with patterns observed earlier in both the DFA analyses and the next-POS probe confusion matrices.

Structurally Stable Classes

These categories remain comparatively stable as context length increases. Their grammatical continuation distributions change more gradually under longer contexts, indicating that they are less sensitive to semantic and discourse-level variation.

They can be characterized by:

- Relatively low JS divergence
- Slower entropy growth
- Higher agreement retention
- Stronger next-POS predictability

Examples of these classes include:

- NOUN
- ADP
- DET
- AUX

These classes correspond closely to the “stable grammatical backbone” identified in the earlier DFA analyses. These are regions of the grammatical state space where continuation behavior remains highly constrained and predictable.

NOUN, in particular, exhibited the lowest divergence from the short-context baseline and consistently maintained the highest agreement rates across context lengths. This suggests that noun continuation behavior forms one of the most stable grammatical structures in autoregressive generation.

Semantically Sensitive Classes

Unlike the structurally stable classes, these categories reorganize much more aggressively as context length increases. These classes appear to be more strongly influenced by discourse structure, topic evolution, and semantic contexts. Their characteristics are:

- Rapid agreement collapse
- Stronger JS divergence growth, and
- Larger entropy swings

Examples of these classes include:

- ADJ
- ADV
- PRON
- PROPN
- VERB

Formatting and Boundary Classes

Categories within this class include:

- SPACE
- PUNCT
- SYM

These classes behaved differently from both the structurally stable and semantically sensitive grammatical categories.

In particular, SPACE exhibited a monotonic decrease in entropy as context length increased. This suggests that formatting and stylistic continuation become increasingly constrained once the model identifies the broader structural and stylistic regime of the document.

Unlike semantic grammatical classes, formatting behavior appears to stabilize under longer contexts because document-level structural regularities become easier to infer over time.

Together, these findings suggest that grammatical continuation behavior is not uniformly sensitive to context. Different POS categories occupy distinct regions of the transformer’s latent

grammatical landscape, each responding differently to increasing semantic and contextual information.

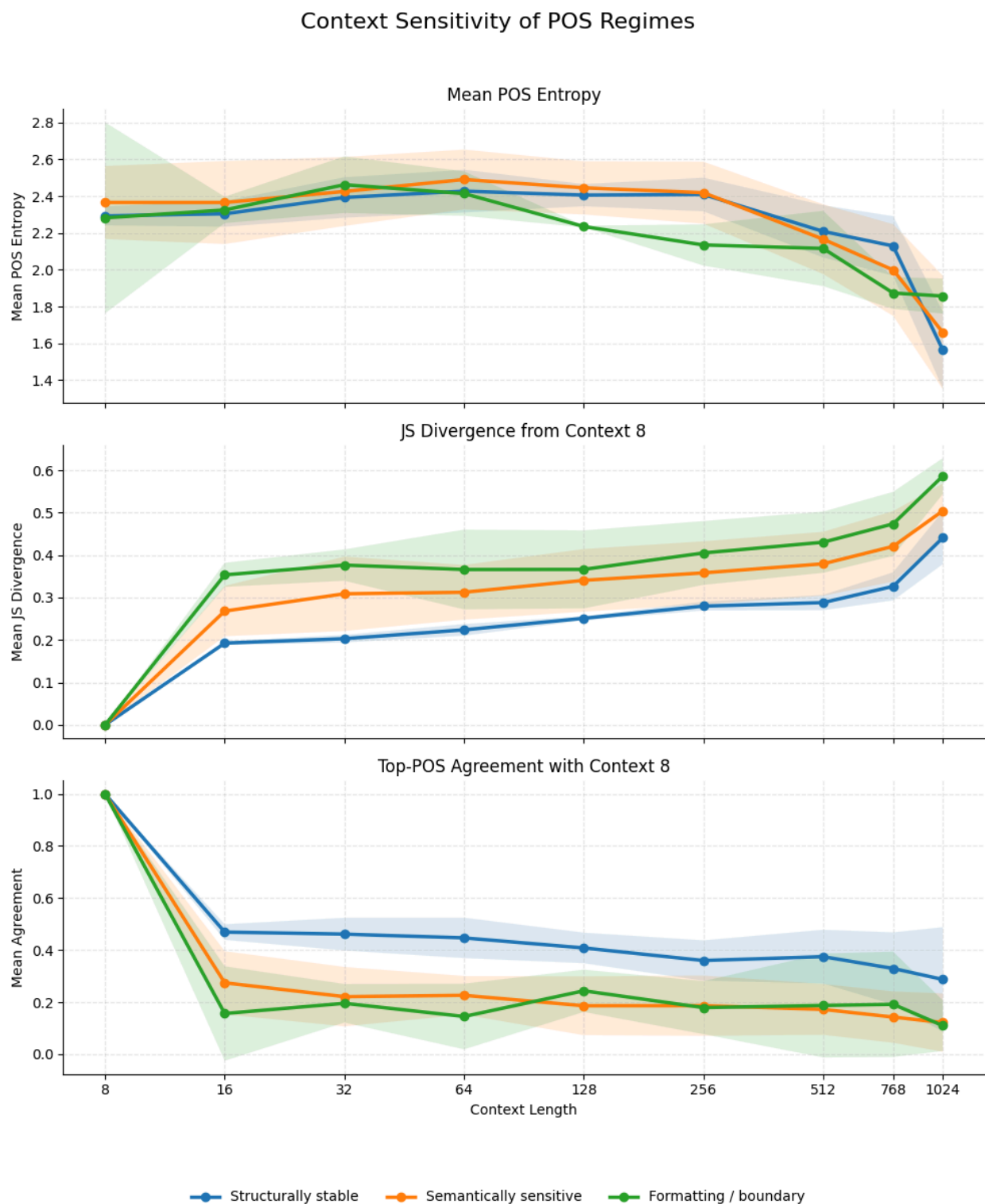


Figure 7: Mean next-POS entropy (top), Jensen–Shannon divergence from the local-context baseline (middle), and top-POS agreement with the local-context baseline (bottom) are shown as

context length increases while the local suffix remains fixed. Curves are aggregated across three groups of POS categories identified in the probe and DFA analyses: structurally stable classes (NOUN, ADP, DET, AUX), semantically sensitive classes (e.g., VERB, ADJ, ADV, PRON, PROPN), and formatting/boundary classes (SPACE, PUNCT, SYM).

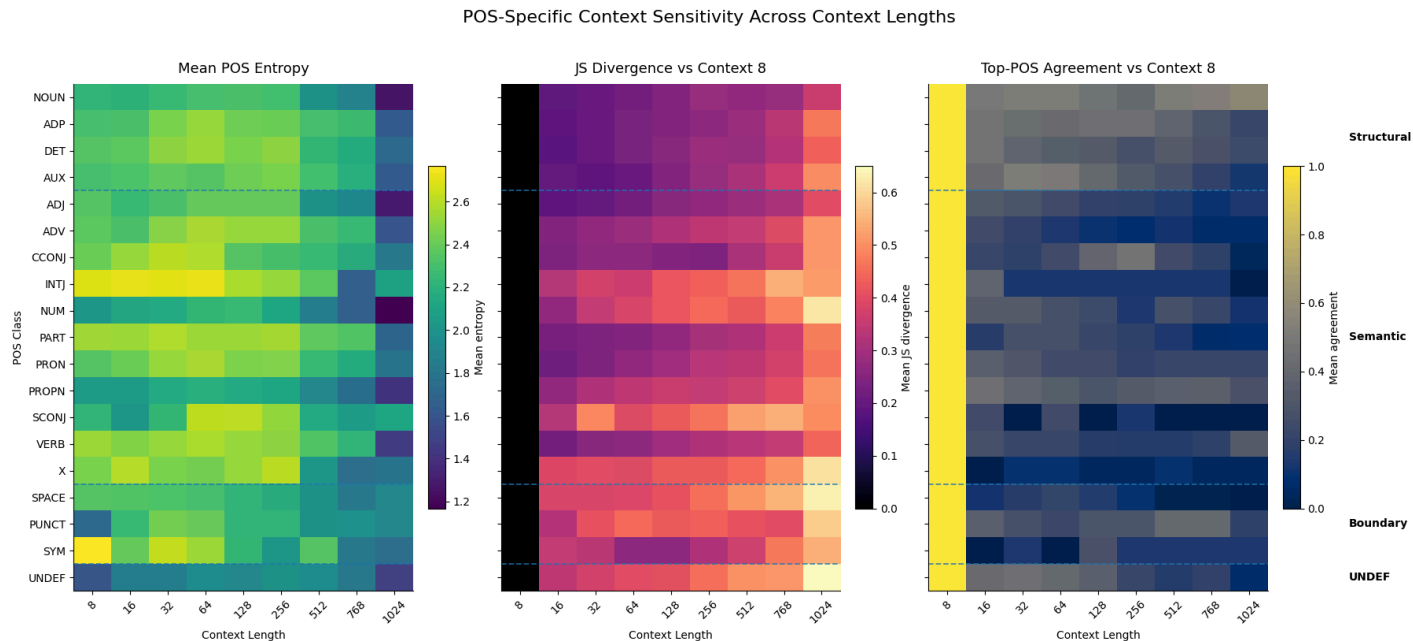


Figure 8: POS context-sensitive heatmaps by mean-next entropy (left), Jensen-Shannon divergence from the local-context baseline (middle), and the top-POS agreement with the local-context baseline (right)

4. Hierarchical Grammar-Aware Generation

The findings from both the DFA-based analyses and the probe and entropy experiments suggest that transformer language generation contains both structured and probabilistic components. The hidden-state trajectory appears to encode grammatical continuation structure in a relatively organized and state-like manner, while much of the uncertainty surrounding token selection emerges later during vocabulary competition and output decoding.

Taken together, these results support the view that autoregressive generation may naturally decompose into two interacting processes:

- a structured latent trajectory governing grammatical continuation behavior,
- and a probabilistic lexical realization process governing token-level competition.

This distinction creates an opportunity to build a hierarchical, grammar-aware generation system that exploits stable grammatical structure to accelerate inference while preserving the expressive power of the full language model when uncertainty becomes high.

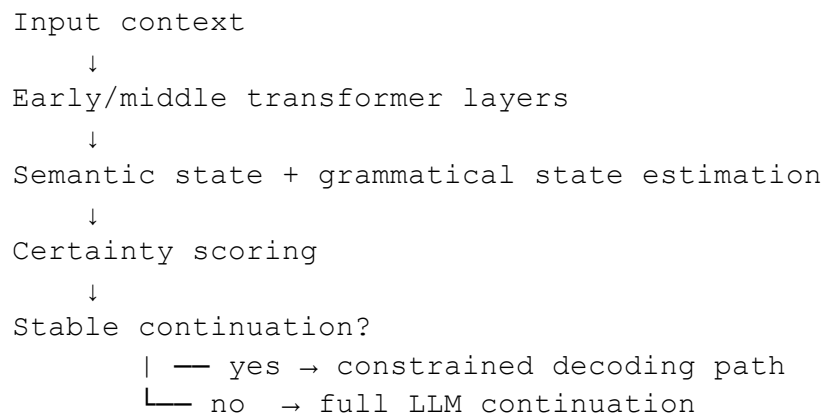
High-Level Architecture

The system consists of five main components:

1. Semantic-to-Grammar Mapper
2. Grammatical State Controller
3. Certainty Estimator and Router
4. Constraint Fusion Layer
5. Constrained Decoder

At each generation step, the system estimates whether the next continuation lies in a stable grammatical region. If the continuation appears structurally stable, the system applies lightweight grammatical constraints to guide token selection using a cheaper decoding path. If uncertainty exceeds a threshold, control is routed back to the full LLM inference path.

Conceptually, execution flow through the system proceeds as follows:



Semantic-to-Grammar Mapper

The semantic-to-grammar mapper connects longer-range semantic context to local grammatical continuation behavior. The system first extracts sequence-level or final-token hidden representations from intermediate layers, most likely layers 4, 5, or 6 based on the probe experiments. These representations are clustered into discrete semantic regions using methods such as vector quantization, product quantization, or k-means clustering.

Each cluster acts as a coarse semantic state representing broader contextual structure within the evolving sequence. For each semantic cluster, the system estimates a conditional distribution over local grammatical continuations:

$P(\text{next_POS} \mid \text{semantic_cluster}, \text{POS_context})$

This distribution models how broader semantic context reshapes local grammatical continuation behavior. The earlier DFA analyses and next-POS probe experiments suggest that local POS context remains the dominant driver of immediate grammatical continuation, while long-range semantic information acts primarily as a distributional modulator. The semantic-to-grammar mapper, therefore, learns how semantic regions influence grammatical branching behavior, continuation stability, uncertainty expansion, and transition preference distributions.

Grammatical State Controller

The grammatical state controller tracks the evolving local grammatical trajectory of the sequence using recent POS continuation history, such as a rolling 5-gram POS context. This component is implemented using the DFA-derived transition models constructed from the earlier grammatical transition analysis. The controller operates as a lightweight local dynamics model that estimates which grammatical continuations are most plausible given the recent POS trajectory.

Rather than using a single monolithic automaton, the system maintains a bank of grammatical transition models, where each automaton corresponds to a target POS class or broader grammatical region. Unlike traditional deterministic finite automata that produce a strict binary accept/reject decision, the controller instead produces soft acceptance scores representing the compatibility of the current POS context with different grammatical continuation classes.

For example:

```
Input POS context:
    DET ADJ NOUN AUX VERB
```

```
DFA bank output:
    NOUN: 0.72
    PUNCT: 0.18
    ADP: 0.07
    VERB: 0.03
    ...
```

This produces a grammatical constraint distribution that is used to adjust token logits during decoding. The resulting grammatical constraint distribution is later used to guide token selection during decoding by reweighting candidate logits, suppressing grammatically implausible continuations, and constraining decoding to structurally stable regions where confidence is sufficiently high.

Constraint Fusion Layer

The grammatical controller does not directly generate lexical tokens. Instead, it constrains or reweights the language model's candidate token distribution using the grammatical continuation

estimates produced by the DFA-based controller. However, the local grammatical controller alone is unlikely to provide sufficient information for accurate constrained decoding in semantically sensitive regions of generation.

The earlier probe and context-sensitivity experiments revealed that some grammatical classes, particularly NOUN, DET, AUX, and ADP remain relatively stable under increasing context and are therefore governed by local grammatical dynamics. In contrast, classes such as VERB, ADJ, ADV, PRON, and PROPN exhibit stronger context sensitivity. As a result, the semantic-to-grammar mapper must actively modulate the local grammatical constraints produced by the DFA controller.

The grammatical controller estimates a local continuation distribution:

$$P_{\text{DFA}}(\text{next_POS} \mid \text{POS_context})$$

While the semantic-to-grammar mapper estimates a semantically conditioned distribution:

$$P_{\text{sem}}(\text{next_POS} \mid \text{semantic_cluster}, \text{POS_context})$$

The constraint fusion layer combines these two signals to produce a final grammatical continuation distribution:

$$P_{\text{final}}(\text{next_POS}) = \text{combine}(P_{\text{DFA}}, P_{\text{sem}})$$

This fusion allows long-range semantic context to reshape local grammatical continuation behavior without completely overriding the structured grammatical backbone captured by the DFA controller.

A simple implementation can be formulated using weighted interpolation:

$$P_{\text{final}} = \alpha \cdot P_{\text{DFA}} + (1 - \alpha) \cdot P_{\text{sem}}$$

where $\alpha \in [0, 1]$ controls the balance between local grammatical stability and semantic contextual modulation. The weighting factor can be estimated empirically using DFA state support counts, semantic cluster support, entropy estimates, agreement statistics, or routing confidence. Highly supported grammatical regions would bias fusion toward the DFA controller, while sparse or semantically unstable regions would increase reliance on the semantic mapper.

Certainty Estimation and Routing

The certainty routing estimator determines whether generation can safely proceed using the cheaper grammar-aware decoding path or whether control should fall back to the full LLM inference path.

The routing decision is based on the estimated stability of the current grammatical continuation region.

At its simplest, a certainty score can be defined as:

$$\text{certainty} = \max P(\text{next_POS} \mid \text{semantic_cluster}, \text{POS_context})$$

where the system estimates the probability of the most likely grammatical continuation given both the current semantic state and the recent local POS trajectory.

Many of the stability signals used by the router are not directly observable during live inference without performing expensive vocabulary-level decoding. Instead, these signals are estimated offline during training and later retrieved or predicted during generation.

Offline Stability Estimation and Live Inference Routing

During offline training, the system collects continuation statistics over large numbers of observed sequences. For each `(semantic_cluster, POS_context)` pair, the system estimates:

- Next-POS distributions,
- Next-POS entropy,
- Top-1 and top-2 POS probability margin,
- The number of plausible POS continuations
- DFA state support counts
- Semantic cluster support counts, and
- UNDEF continuation frequency

These statistics provide empirical estimates of how stable or ambiguous a continuation region tends to be. During live inference, the system does not directly compute full-next token distributions before routing. Instead, the router uses either cached empirical statistics, learned router predictions, or a hybrid combination of both. This allows the system to make routing decisions without requiring full vocabulary-level inference at every generation step.

Stable vs Unstable Continuation Regions

A continuation is considered stable when:

- one grammatical continuation strongly dominates,
- entropy remains low,
- the margin between competing POS classes is large,
- the local POS context has strong DFA support,
- and the semantic cluster contains sufficient historical continuation statistics

Conversely, unstable regions emerge when:

- multiple grammatical continuations compete,
- entropy increases,
- semantic conditioning reshapes the continuation distribution,
- or the observed POS trajectory falls into sparsely supported transition regions.

The router attempts to estimate whether the current continuation lies within stable grammatical structure or an ambiguous continuation space. Given a routing threshold, the router then determines whether to use the constrained decoding path or defer to the full language model. High certainty continuations are delegated to the constrained grammar-aware decoder, and low certainty regions are handled by the full LLM inferencing process.

The constraint fusion layer could also provide an additional uncertainty signal for the routing mechanism. If the semantic mapper and grammatical controller strongly disagree about the next grammatical continuation, this disagreement itself may indicate semantic ambiguity, grammatical instability, or insufficient local support. Such disagreement may also trigger a fallback to the full LLM decoding path.

Constrained Decoder Mechanism

A simple constrained decoding formulation can be written as:

$$\text{adj_logit}(\text{token}) = \text{orig_logit}(\text{token}) + \lambda \cdot \Sigma_{\text{pos}} \cdot \text{fusion_score}(\text{pos}) \cdot P(\text{pos} \mid \text{token})$$

Where:

- $\text{orig_logit}(\text{token})$ comes from the LM head,
- $\text{fusion_score}(\text{pos})$ is the semantically modulated grammatical continuation score produced by the constraint fusion layer,
- $P(\text{pos} \mid \text{token})$ is a POS compatibility distribution associated with the candidate token,
- and λ controls the strength of the grammatical constraint.

Rather than assigning each token a single deterministic POS label, the decoder estimates a soft POS compatibility distribution over the vocabulary.

$$\text{POS}(\text{token}) = P(\text{POS} \mid \text{token_id})$$

This distribution captures the fact that many lexical items are grammatically ambiguous and may function as different POS classes depending on context. The compatibility distribution can be estimated using embeddings, corpus statistics, or lightweight contextual probe predictors. Using soft compatibility scores allows the system to bias generation toward grammatically plausible continuations while still preserving lexical flexibility within each grammatical category.

However, the earlier probe experiments revealed an important architectural challenge. Although grammatical continuation structure is highly recoverable from intermediate transformer layers, the standard LM head is trained only on the final hidden representation. As a result, raw layer 4–6 hidden states are unlikely to produce well-calibrated vocabulary distributions without additional transformation.

To address this, two possible constrained decoding strategies for enabling early grammar-aware token generation from intermediate layers will be explored.

Option A: Early-Exit LM Head

The first approach attaches a lightweight vocabulary projection head to an intermediate transformer layer, such as layer 6.

```
h_6
  ↓
Early-exit LM head
  ↓
Candidate token logits
  ↓
POS-constrained reweighting
  ↓
Token selection
```

The early-exit head is trained using knowledge distillation to imitate the final GPT-2 output distribution. A possible training objective is

```
L = KL(softmax(final_logits) || softmax(early_logits))
```

where `final_logits` are produced by the full GPT-2 model and `early_logits` are produced directly from the intermediate hidden representation. The objective is not necessarily to perfectly reconstruct the full vocabulary distribution, but rather to recover enough of the dominant top-k candidate structure to support constrained decoding in stable grammatical regions. This approach is computationally simple and directly tests whether middle-layer grammatical representations contain sufficient information for accelerated generation.

Option B: Bridge Transformation Network

The second approach learns a lightweight transformation that maps an intermediate hidden representation into an approximation of the final hidden representation before applying the original LM head.

```
h_6
  ↓
Bridge transform
```

↓
approximate h_{12}
↓
Original LM head
↓
Candidate token logits

Instead of training a new LM head, this approach attempts to preserve compatibility with the existing GPT-2 output geometry.

A possible training objective is

```
L = MSE( bridge(h_6), h_12 )
```

A stronger objective combines hidden-state reconstruction with logit matching:

```
L = MSE(bridge(h_6), h_12) + KL(final_logits || bridge_logits)
```

```
where bridge_logits = LM_head(bridge(h_6))
```

This approach is more architecturally elegant because it preserves the original LM head but may be harder to optimize than the early-exit head.

Two-Timescale Generation and Cache Synchronization

The proposed architecture operates across two interacting timescales: a slow semantic state and a fast grammatical state.

The slow semantic state evolves gradually and is updated periodically using intermediate hidden representations or semantic cluster assignments. Managed indirectly by the semantic-to-grammar mapper, this state captures broader sequence-level structure, including topic evolution, discourse organization, narrative progression, and long-range contextual dependencies. Conceptually, it acts as a coarse latent trajectory governing the broader direction of generation.

The fast grammatical state evolves at the token level and is updated after each generated token or completed word. It tracks the local POS trajectory, DFA transition state, grammatical continuation constraints, and short-range syntactic structure. The grammatical controller therefore acts as a lightweight local dynamics model responsible for immediate continuation behavior.

Together, the semantic-to-grammar mapper and grammatical state controller create a hierarchical generation system in which semantic evolution and grammatical evolution operate as partially decoupled processes across different temporal scales.

Once the constrained decoding path begins generating tokens independently of the full transformer stack, an important engineering challenge emerges. If portions of the transformer are bypassed during grammar-aware generation, the hidden state and KV cache will eventually become stale relative to the generated sequence. To address this, periodically generated tokens must be passed back through the full transformer to refresh hidden representations, semantic cluster assignments, and KV-cache state before generation can safely continue using the full LLM path. This creates a central systems-level tradeoff. More constrained decoding steps mean greater computational savings but also greater semantic-state drift risk.

The earlier experiments suggest that grammatical continuation structure remains relatively stable over short horizons, while semantic structure evolves more gradually. This raises the possibility that multiple local grammatical continuation steps can be generated using the fast grammatical controller before recomputing the slower semantic state. Future experiments will investigate:

- semantic-state drift over constrained generation horizons,
- degradation in grammatical continuation quality over time,
- synchronization frequency requirements,
- and the tradeoff between computational savings and semantic fidelity.